

GIRIJANANDA CHOWDHURY UNIVERSITY, GUWAHATI
DEPARTMENT OF CIVIL ENGINEERING

Formative Assessment - II

Geotechnical Engineering - BCE23403T (B. Tech 4th Semester)

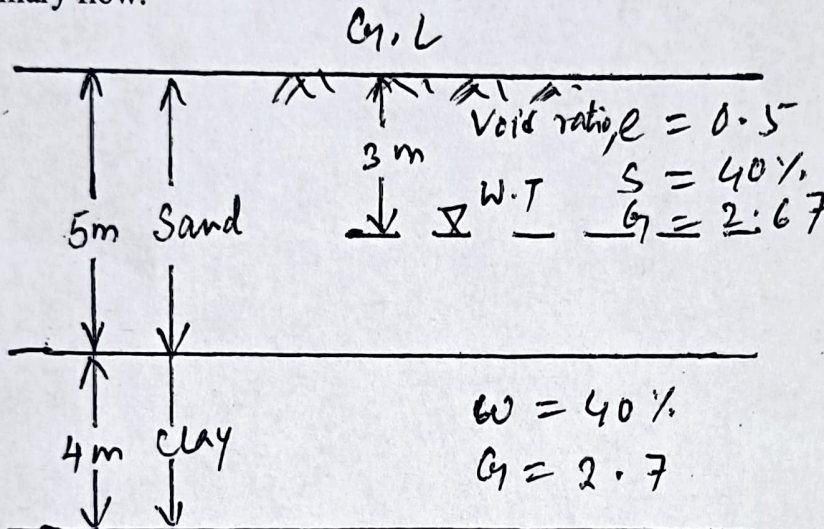
Batch: 2023-2027

22nd April 2025

Academic Year: 2024-2025

Total Marks: 30

Duration: 1 hour

Questions	Marks	CO
1 a) What is quick sand condition? b) What is zero air void line? c) What is primary and secondary consolidation? d) Define contact pressure of soil for rigid and flexible footing.	3 x 4 =12	1
2 For the subsoil condition shown in figure below draw the total, neutral and effective stress diagrams up to a depth of 9 m, neglecting capillary flow.  Fig:- Soil Profile.	6	2
3 A normally consolidated clay layer settled by 30 mm when the effective stress was increased from 35 to 60 kN/m². What will be its settlement when the effective stress is increased from 60 to 110 kN/m²?	6	2
4 Find the intensity of vertical stress and horizontal shear stress at a point 3 m below and 1 m horizontally away from the axis of loading of a 30 kN point load acting at the horizontal ground surface.	6	2